

Muhammad Rafi

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Education

Presidency University

Bangalore,India

B.Tech, Computer Science and Engineering (GPA: 8.37)

Experience

VirtualStaff

Jun 2024 - Present

Software Engineer

Philippine (Remote)

- Developed the Seat System from scratch, integrated Stripe payments, and shipped it to production, generating \$50k+ in revenue.
- Built an algorithm to calculate guaranteed staff payouts, handling USD-to-PHP conversion, FX rates, and automated invoice calculations.
- Optimized backend APIs and database queries to support 1M+ users, improving performance across high-traffic production systems.
- Resolved production issues reported by customer support teams, debugging user-facing bugs and shipping fixes across frontend and backend systems.

Indian Space Research Organisation

Jan 2024 - April 2024

Unity Developer

Bangalore,India

- Collaborated closely with ISRO's astronaut training experts, resulting in a simulator that closely mirrors the challenges of real-world space missions,docking.
- Engineered a highly immersive astronaut training experience by employing Unity and pioneering 6 Degrees of Freedom (6DoF) technology, resulting in a 30% increase in realism and user engagement.
- Executed UDP for efficient communication, significantly reducing latency and enhancing real-time responsiveness by 20% (27ms)

Quantum Strategies

June 2023-June 2024

Software Engineer

Pennsylvania,United States (Remote)

- Built the backend foundation from scratch at an early-stage startup, implementing Hexagonal Architecture with Node.js and Express for clean service separation.
- Collaborated with the frontend developer to seamlessly integrate user interfaces with backend functionalities, enhancing overall user experience.
- Designed and implemented CI/CD pipelines to automate testing and deployment of backend APIs, accelerating release cycles and ensuring high code quality for application development.

Freelance

Mar 2020-June 2023

Full Stack Developer,UI-UX Designer

- Built engaging web platforms, including a 3D home builder showcase and a redesigned hotel menu, improving user interaction and boosting engagement by 20%.
- Built RentQuicker, a full-stack rental platform with property listings, search filters, booking flows, PPP-based regional pricing, and Stripe/PayPal payment integrations.

Technical Skills

Languages: JavaScript, TypeScript, Java, Python, C, C++, Go, Rust

Frameworks: Node.js, Express.js, React.js, Next.js, Angular, Spring Boot, FastAPI, SASS, Axum, Actix-web

AI/LLM: LangChain, LangGraph, OpenAI API, Claude API, RAG, Vector Databases, Embeddings, AI Agents, MCP, Function Calling, Prompt Engineering

Databases: MongoDB, MySQL, PostgreSQL, Redis, Pinecone, ChromaDB

Cloud/DevOps: Docker, Kubernetes, AWS, Oracle Cloud, CI/CD, GitHub Actions, Nginx

Backend: REST APIs, WebSockets, Microservices, Message Queues, BullMQ, Kafka, Pub/Sub, Authentication, Payments

Concepts: Data Structures, System Design, Distributed Systems, Database Design, Operating Systems, Machine Learning, Agile